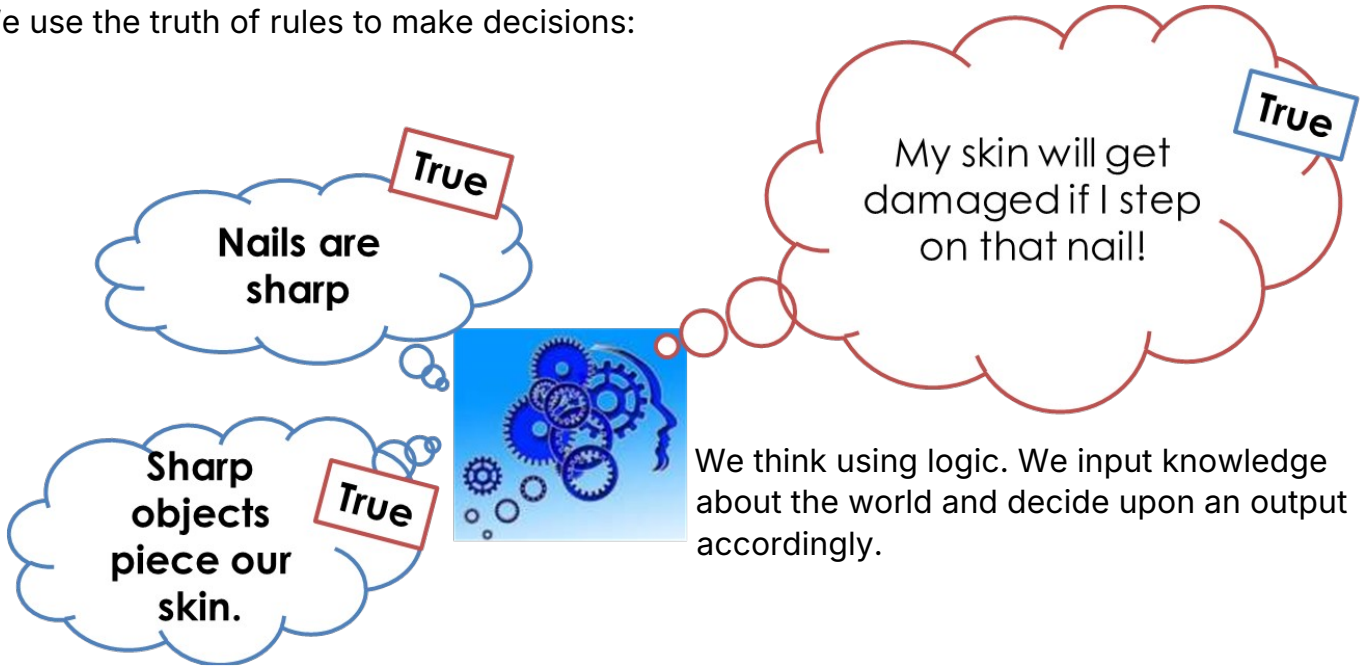
Computational Logic

Introduction – What is Logic?

Logic is used to make decisions. Generally speaking, the decisions we make are determined by rules that we know about the world. The decisions we make are based on whether or not these rules are true or false in a given situation. For example, if we see a nail sticking up from a piece of wood on the ground, we will decide to avoid it. This is because we know, as a rule, that nails are sharp AND that sharp objects pierce our skin.

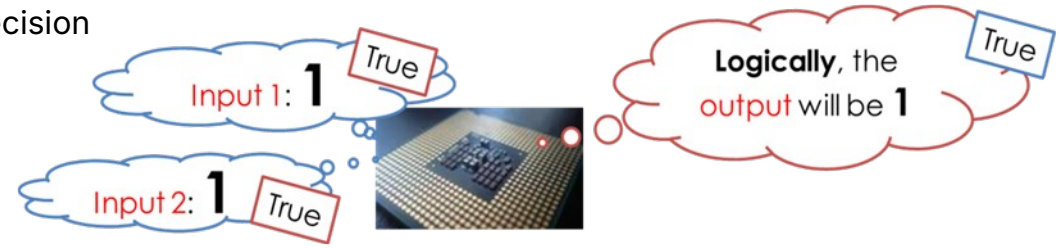
We use the truth of rules to make decisions:



We think using logic. We input knowledge about the world and decide upon an output accordingly.

What is Binary Logic?

Human logic works with rules / facts / statements which are either True or False. Computers have been designed to do the same. Binary logic is what a CPU uses to make decisions based on the inputs it receives. The CPU (like us) will take inputs which may be true or false (1 or 0) and make a decision which will produce a particular output.



There are very few logic operations that a CPU will perform on its inputs. However, combinations of these lead to incredibly complex calculations. The logic operations that a CPU will perform are:

The **NOT** operation

The **AND** operation

The **OR** operation

Electronics engineers have developed components that can carry out the AND, OR and NOT logic operations. The CPU is full of millions of these components. These components are

Computer Science UK Membership Site Licence: Do not share outside your centre.

known as 'logic gates'. They are made up of transistor switches arranged in particular ways. Because they are so small, we use symbols to represent the 3 logic gates.

Invert or NOT operation

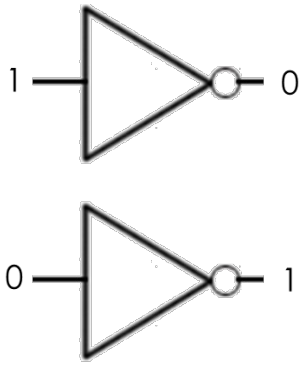
The NOT gate produces an output which is the reverse of the input.

If the input is 1 (Logic True) then the output is 0 (Logic False).

If the input is 0 (Logic False) then the output is 1 (Logic True)

The shorthand way of writing this is Q = NOT A (where 'A' is the input and 'Q' is the output).

It does not have to be A and Q, you could use any letter, but Q is a popular choice to describe a logic output.

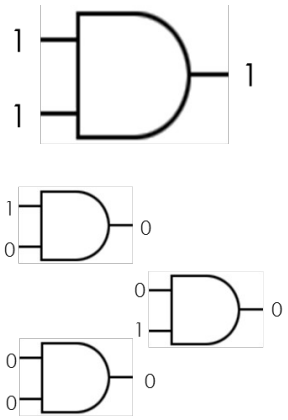


The AND operation

The AND gate takes two inputs, to produce a single output.

The rule for this gate is that if both A and B are True then the output is also True, otherwise it is False.

The shorthand for this is Q = A AND B

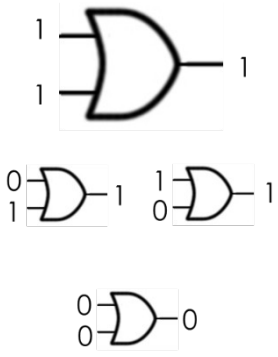


The OR operation

The AND gate takes two inputs, to produce a single output.

The rule for this gate is that if either input, or both inputs are True then the output is also True.

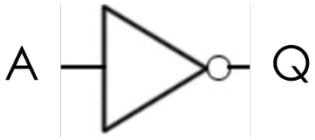
The shorthand for this is Q = A OR B



Truth Tables

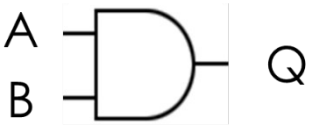
Computer Science UK Membership Site Licence: Do not share outside your centre.

Truth tables are used to work out the different combinations of inputs that a given logic gate could receive and the resulting outputs that are created from its operation.



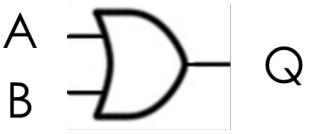
NOT Gate Truth Table:

A	Q
1	0
0	1



AND Gate Truth Table:

A	B	Q
1	1	1
1	0	0
0	1	0
0	0	0



OR Gate Truth Table:

A	B	Q
1	1	1
1	0	1
0	1	1
0	0	0

Binary Logic Circuits

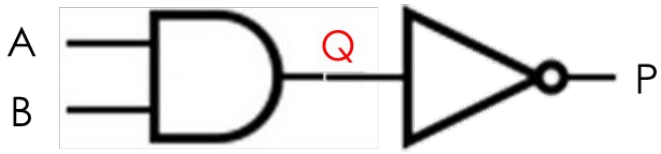
Computer Science UK Membership Site Licence: Do not share outside your centre.

In your exam you will be expected to be able to look at a circuit and produce a truth table for the inputs and outputs.

Tips:

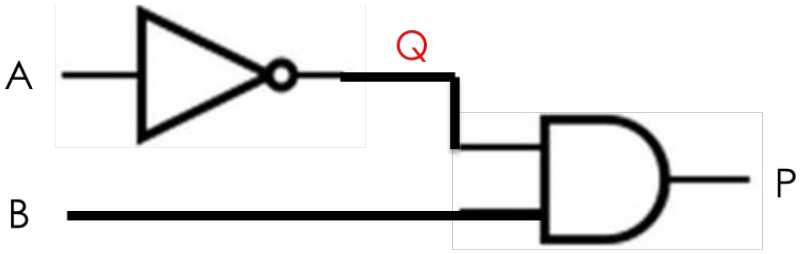
- Label the inputs on the circuit diagram
- Label the outputs on the circuit diagram
- Create a table with a column for each input and output
- Label the columns (outputs labelled in the form of an expression)
- Place the combinations of different zeros and ones in the input columns
- Workout the outputs for each output column in turn

Example 1:



A	B	Q = A AND B	P = NOT (A AND B)
1	1	1	0
1	0	0	1
0	1	0	1
0	0	0	1

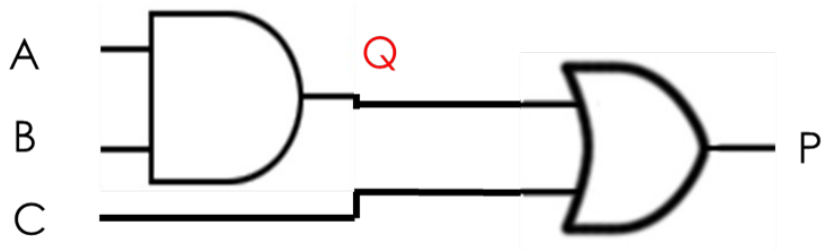
Example 2:



A	B	Q = Not A	P = (NOT A) AND B
1	1	0	0
1	0	0	0
0	1	1	1
0	0	1	0

Computer Science UK Membership Site Licence: Do not share outside your centre.

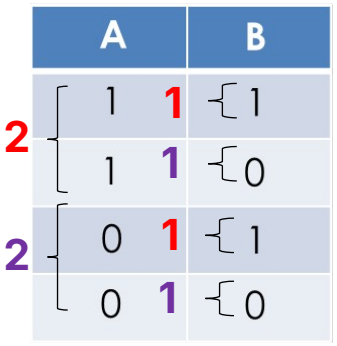
Example 3:



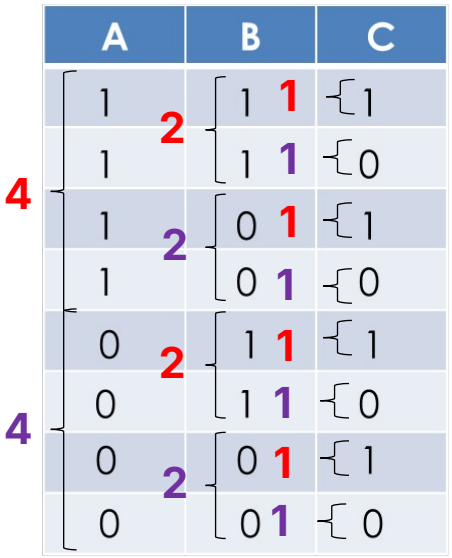
A	B	C	Q = A AND B	P = (A AND B) OR C
1	1	1	1	1
1	1	0	1	1
1	0	1	0	1
1	0	0	0	0
0	1	1	0	1
0	1	0	0	0
0	0	1	0	1
0	0	0	0	0

More Tips:

When you place the combinations of different zeros and ones in the input columns, there are patterns that you can remember, to help you:



2 inputs



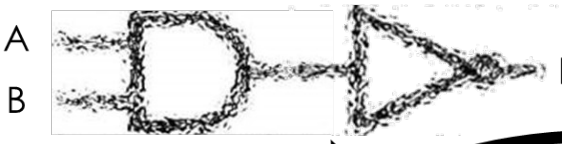
3 inputs

What if there were 4 inputs?

Computer Science UK Membership Site Licence: Do not share outside your centre.

Binary Logic Circuit Expressions

In your exam you will also be expected to be able to draw a diagram from an expression.



How to we draw a diagram from an expression

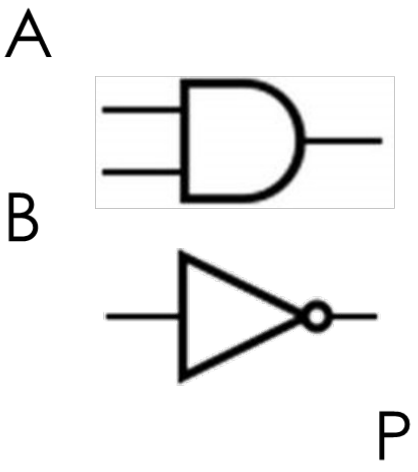
A	B		P = NOT (A AND B)
1	1	1	0
1	0	0	1
0	1	0	1
0	0	0	1

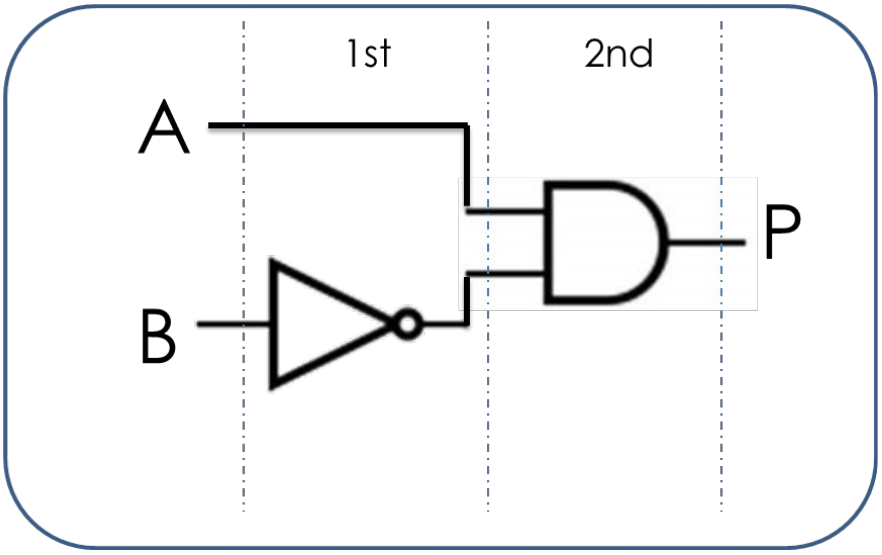
Example 1:

- Write down the Inputs, Gates and Outputs of the expression.
- Consider how many parts there are to the expression (in this one there is the 'bracket' part and the 'outside bracket' part – therefore 2 parts to the expression) and split your drawing space into that number of parts.
- Place the inputs to the left of the first space and the output to the right of the second space.
- Place the gate that is inside the bracket into the first space
- Place the gate that is outside the bracket into the second space.
- Join up the inputs and outputs logically.

P = **A** AND (NOT **B**)

----BRACKETS FIRST!----



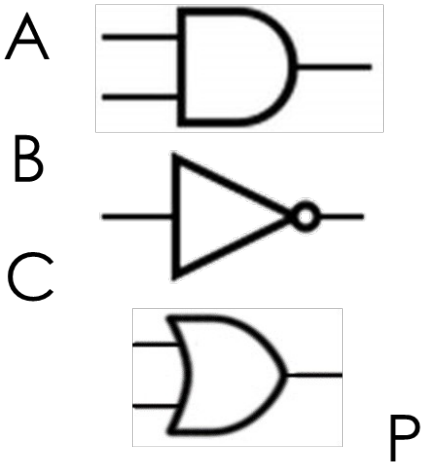


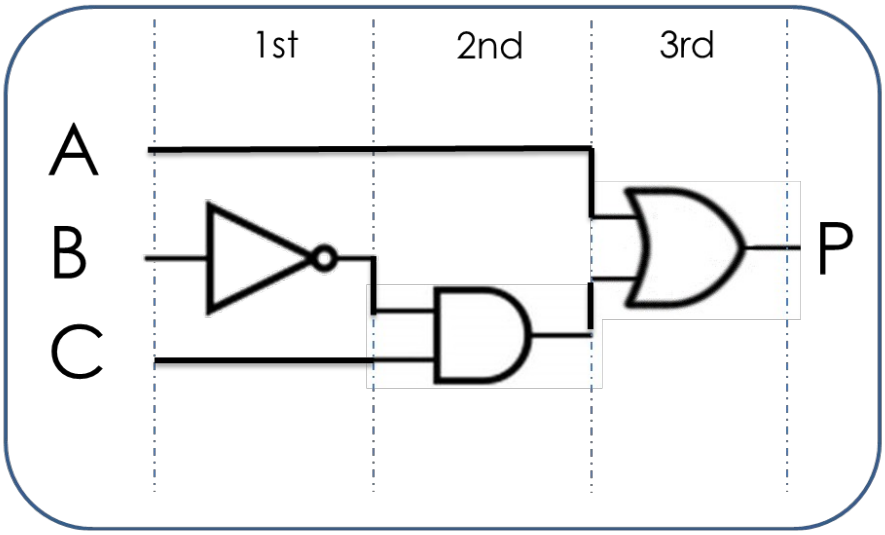
Example 2:

- Write down the Inputs, Gates and Outputs of the expression.
- Consider how many parts there are to the expression (in this one there is the 'brackets inside brackets' part, a 'bracket' part and the 'outside bracket' part – therefore 3 parts to the expression) and split your drawing space into that number of parts.
- Place the inputs to the left of the first space and the output to the right of the second space.
- Place the gate that is inside the brackets in brackets, into the first space
- Place the gate that is inside the brackets into the second space
- Place the gate that is outside the bracket into the third space.
- Join up the inputs and outputs logically.

P = ((NOT **B)** AND **C**) OR **A**

----BRACKETS inside BRACKETS FIRST!----





Questions *(The question zone you choose must either match your target grade or be higher!)*

Question Zone 1-3

1	2	3	4	5	6	7	8	9

1. Draw and label the OR, AND & NOT logic gates. [3]
2. Create and complete truth tables for each of the logic gates. [3]
3. Create and complete the truth table for the logic circuit with the expression: [4]

Q = A AND (NOT B)

A	B	NOT B	A AND (NOT B)

Question Zone 4-6

1	2	3	4	5	6	7	8	9

1. Create and complete the truth table for the logic circuit with the expression: [4]

Q = A AND (NOT B)

A	B	NOT B	A AND (NOT B)

2. Draw the logic circuit diagrams from the following expressions:
   a) Q = NOT (A AND B) [1]
   b) Q = NOT (A OR B) [1]
   c) Q = (NOT A) OR (NOT B) [1]
   d) Q = A AND (NOT B) [1]
   e) Q = (NOT A) AND B [1]

3. Create and complete the truth table for the logic circuit with the expression: [4]

Q = (A OR B) AND C

Questions Zone 7-9

1	2	3	4	5	6	7	8	9

1. Create and complete the truth table for the logic circuit with the expression: [4]

Q = (A OR B) AND C

2. Draw the logic circuit diagrams from the following expressions:
   a) Q = (A AND B) OR C [1]
   b) Q = A AND (B OR C) [1]
   c) Q = (A OR B) AND (NOT C) [1]
   d) Q= NOT ((A AND B) AND C) [1]
   e) Q = (A AND B) OR (A AND C) [1]

3. Draw the logic circuit diagrams from the following expressions:

Q = NOT ((A AND B) OR NOT (C AND D)) [2]

Computer Science UK Membership Site Licence: Do not share outside your centre.

Checklist:

- Date and title, clearly presented
- Spelling & grammar checked
- Question numbers in the margin
- Handwriting neat & legible
- Punctuation / Capital letters

Keywords / Key Terms:

Binary Logic: Also known as Boolean Logic, it is the process of reducing values to either True or False, in order to perform calculations

Logic Gate: A logic gate is a building block of the digital circuitry found in a CPU. It performs a logical operation on the inputs it receives.

Truth Table*:* A table showing all possible outputs from the various combinations of inputs that a gate can be supplied with.

Computer Science UK Membership Site Licence: Do not share outside your centre.

Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
Explain: Explaining is 'saying WHY/HOW something is like that'. (E.G.: A computer will have a CPU so that it can process all of the data the computer needs to perform a range of tasks. Primary and Secondary storage is needed because...)
Discuss: Discussing is 'looking at two sides of an issue, weighing up the two views and giving a conclusion'. Often these require a mini essay answer. (E.G.: New technology could be seen as being bad for the environment because..., but on the other hand, new technology has led to... In conclusion I believe that...)
Describe/Explain/Discuss **using examples**: Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**!

Computer Science UK Membership Site Licence: Do not share outside your centre.

Reflections **Score:** / **Percentage:** % **Grade:** **Progress** On / Above / Below

What Went

- ☐ I demonstrated a good level of understanding.
- ☐ I responded to the command words effectively.
- ☐ My answers were detailed / were written in depth.
- ☐ My work was well presented / legible.
- ☐ My answers effectively incorporated technical terminology.
- ☐ My responses were well structured / organised.
- ☐ My revision strategy was effective as I showed depth of understanding in my answers.
- ☐ My answers contained enough points / examples / explanations to achieve the marks available.

Even Better

- ☐ My answers need to be more accurate.
- ☐ I must respond correctly to the command words.
- ☐ My answers need more detail / greater depth.
- ☐ I must take greater care over my work / write neatly.
- ☐ I must incorporate key terminology into my answers.
- ☐ I must better organise my answers to improve its clarity.
- ☐ I need to improve my revision strategy, as I did not demonstrate a depth of understanding in my answers.
- ☐ My answers didn't contain enough points / examples / explanations to achieve the marks available.

Further thoughts:

Computer Science UK Membership Site Licence: Do not share outside your centre.